2017 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.201 Databases

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Question 1

1.a Normalisation, within the context of databases, aims to remove dependencies that can be the source of operational problems. Explain this statement giving examples to illustrate your answer.

[3 marks]

1.b Given the ProjectHours relation *on the following page*:

i. Explain why the relation is in first normal form

[2 marks]

ii. Identify the functional dependencies that exist in the relation being careful to identify the type of each

[5 marks]

iii. With reference to the dependencies you have identified, normalise the relation into second normal form

[3 marks]

iv. State whether the design in second normal form is also in third normal form (3NF), and

- If it is, outline why this is the case using the definition of 3NF as the basis for your argument
- If it is not, identify what is still wrong and continue the normalisation process to give a solution in 3NF

[3 marks]

1.c Being careful to declare any keys, provide SQL statements that would create the final set of tables from part b.

[9 marks]

Question 1 continues on next page...

Question 1 continued...

ProjectHours relation for Question 1 part b

StaffID	StaffName	Department	Project	ClientID	ClientName	Task	Hours
801	Jones	Engineering	Tesla	45	Tesla Inc	1	12
948	Sun	Engineering	Tesla	45	Tesla Inc	1	96
831	Walker	Physics	CPower	56	TransPower	1	8
894	Chapman	Computing	MANET	72	Acme	1	22
873	Allen	Computing	Cerberus	98	XY Robotics	1	120
802	Peterson	Engineering	Cerberus	98	XY Robotics	1	89
801	Jones	Engineering	Tesla	45	Tesla Inc	2	48
948	Sun	Engineering	Tesla	45	Tesla Inc	2	48

Total 25 marks

Question 2

You have been tasked with producing a database for a charity that restores and operates trains. The charity largely works on an ad hoc basis but they need to keep better track of their volunteers and trains.

Railway locomotives are stored in one of a number of named depots, each having a single manager. It would be useful to have a record of the name, phone number, and capacity (number of locomotives) for each depot.

Managers and other individuals volunteer at one or more depots and to assist with operations the charity would like to have their name, address, and phone number on file.

The charity would like to keep track of the number of hours per week that the volunteers are normally willing to work – the charity doesn't keep track of the number of hours volunteers actually work as the hours can vary considerably and are organised ad hoc.

Locomotives have a unique registration number and are owned by one or more of the managers or volunteers; due to their cost locomotives will typically have shared ownership. For legal reasons the charity needs to keep a record of the owners of each locomotive and the date at which it was purchased by them. They also need to record the date at which the locomotive last had its safety certificate issued.

As locomotives are purchased by volunteers rather than the charity it is common to have multiple locomotives of the same class. To help with the operational side of their work the charity needs to record the characteristics of each class, including the class number, the power source (whether it is steam, diesel, or electric), the power output of the engine, and its weight.

Finally, the charity needs to record who is licensed to operate each class of locomotive.

2.a Produce an Entity-Relationship (ER) diagram that represents the information needed by the charity.

[15 marks]

2.b Convert your ER model into a set of relations being careful to identify **each and every** step in the process, and the result of applying each step. If a step is not applicable, this must be stated.

[10 marks]

Total 25 marks

Question 3

3.a Briefly describe each of the four properties of an ACID database transaction.

[4 marks]

3.b Consider the pseudo-code below for an `unlock_item` function in a *Binary Lock*. Write pseudo-code for the accompanying `lock_item` function.

```
unlock item(X)

LOCK(X) ← 0

if (transactions are waiting):

  wakeup one of the waiting transactions;
```

[4 marks]

3.c Explain the role of a commit point in the context of a write to a database. Include in your explanation an outline of how they can be used to ensure database integrity. Provide a transactional state diagram to assist your explanation.

[7 marks]

3.d Describe the situation in which a deadlock may occur.

[2 marks]

Question 3 continues on next page...

Question 3 continued...

3.e

i. Consider the pseudo-code below for two transactions. Using the additional available methods provided, modify these transactions to apply *Two Phase Locking.*

Transaction 1

```
read_item(X);

X := X + Y;

write_item(X);
```

Transaction 2

```
read_item(Y);

Y := X + Y;

write_item(Y);
```

Available Methods

```
read_lock( );

write_lock( );

unlock( );
```

[2 marks]

ii. Discuss the different phases of *Two Phase Locking*. Include an explanation of how they guarantee serialisability when used together.

[6 marks]

Total 25 marks

Question 4

4.a Recall and briefly describe two methods of collision resolution used in *Internal Hashing*.

[4 marks]

4.b

i. Explain the concept of buckets in the context of *External Hashing*. Include details of their specific role, especially in their use for record deletion.

[6 marks]

ii. Discuss *Static Hashing* in the context of External *Hashing*; include the impact of changing the number of both records and buckets.

[6 marks]

Question 4 continues on next page...

Question 4 continued...

4.c

i. Describe two positive aspects and one negative aspect of Extendible Hashing.

[3 marks]

ii. Consider the data and current state of the database shown below. Apply *Extendible Hashing* techniques to show how the third record can be added to the hash file. Show each step in detail.

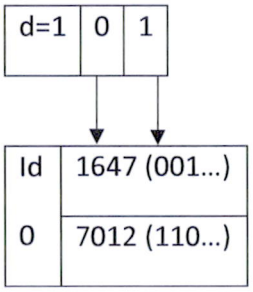
Note that, as our hash function, we take the 13 Least Significant Bit (LSB) binary representation as input. This representation is given in the *Binary* column in the *Data* table below. Also, each data page can hold two records. Use diagrams to aid your explanation of this process.

[6 marks]

Data

Decimal	**Binary**
1647	0011001101111
7012	1101101100100
1734	0011010111100

Current State



Total 25 marks

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